

MAJOR PROJECT REPORT ON

AI-Based Digital Forensics and Cyber Incident Reconstruction

Submitted By

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SEPTEMBER 2026

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In partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

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C E R T I F I C A T E

This is to certify that the project report titled **AI-BASED DIGITAL FORENSICS AND CYBER INCIDENT RECONSTRUCTION** is a bonafide record of the **CSD 415 Project Phase I** work presented by **Asif Subair** (Reg. No. CEC23CS049), **Fidha Fathima V.A** (Reg. No. CEC23CS065), **Mohammed Ancil M.N** (Reg. No. CEC23CS083), and **Nandhana Sunil Deth** (Reg. No. CEC23CS091), Seventh Semester B.Tech. Computer Science and Engineering students, under our guidance and supervision, in partial fulfillment of the requirements for the award of the degree , **Bachelor of Technology in Computer Science and Engineering** of **APJ Abdul Kalam Technological University**.

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ABSTRACT

Cyber incidents generate large volumes of heterogeneous logs, making manual investigation and event correlation difficult. This project proposes an AI-based platform for digital forensic investigation and cyber incident reconstruction. The system is intended to collect authentication, system, network, file, and process logs, normalize their contents, and extract behavioural and temporal features. AI-assisted analysis will identify anomalous activity and support the categorization of events as normal, suspicious, or potentially malicious. The selected base paper, LogDLR, provides a foundation for cross-system log anomaly detection through sentence embeddings, a Transformer autoencoder, and domain-adversarial learning. The proposed project extends this detection perspective by correlating related events across users, devices, IP addresses, resources, and time to reconstruct a probable incident sequence. Its planned outputs include an evidence-linked timeline, an attack-path graph, and risk and confidence indicators presented through an investigation dashboard. Anomalous activity will be treated as an investigation lead rather than proof of an attack, and reconstructed sequences will remain subject to human validation. The objective is to help investigators understand how suspicious events may be connected while retaining access to the supporting evidence. This report presents the proposed approach and literature review; it does not claim completed implementation or measured project performance.

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Chapter 1

INTRODUCTION

Digital forensic investigation examines digital records to understand events associated with a suspected incident. Authentication services, operating systems, network components, files, and processes can produce separate streams of evidence, while an investigator needs a coherent account of what happened. The proposed project addresses the difficulty of examining these heterogeneous records and connecting related activities across entities and time.

The selected base paper, LogDLR: Unsupervised Cross-System Log Anomaly Detection Through Domain-Invariant Latent Representation, addresses anomaly detection when log formats and patterns differ between systems [1]. LogDLR represents log entries using Sentence-BERT (SBERT) embeddings and learns normal sequences using a Transformer-based autoencoder. Domain-adversarial training aligns source- and target-system representations, and reconstruction errors identify anomalous sequences. This provides a foundation for studying heterogeneous log analysis without relying on log-template parsing.

The reviewed literature explores complementary approaches. LogGraph and SwissLog investigate event relationships and anomaly diagnosis or localization [2, 3]. LogRep and DeepUserLog incorporate semantic information, numeric values, and key-value data [4, 5]. Hierarchical Transformers, DeepSyslog, LogBERT, and NeuralLog examine contextual or semantic representations [6–9]. LogTAD studies cross-system transfer through domain adaptation [10]. These methods inform the proposed detection component, but anomaly-detection results should not be interpreted as evidence of complete cyber incident reconstruction.

The project, **AI-Based Digital Forensics and Cyber Incident Reconstruction**, will combine anomaly detection with event correlation and temporal analysis. Shared users, devices, IP addresses, resources, and timestamps will provide candidate links between records. These links will be presented alongside their supporting evidence rather than treated as automatically established causal relationships.

The planned outputs are an evidence-linked timeline, an attack-path graph, and risk and confidence indicators in an investigation dashboard. The attack-path graph will represent a proposed interpretation of related events, not a verified account of attacker actions. Final conclusions will require investigator review because unusual events may also reflect faults, administrative actions, or incomplete data.

Chapter 2

PROBLEM STATEMENT

2.1 Problem Statement

Cyber incidents generate large amounts of log data from different sources. Differences in structure, event descriptions, and timestamps make manual examination and correlation difficult. An investigator may identify a suspicious login, a process event, or a network connection without immediately understanding whether the records belong to the same incident.

The reviewed methods primarily address anomaly detection, diagnosis, localization, or cross-system transfer. These capabilities are useful, but do not by themselves provide the complete evidence-linked reconstruction workflow proposed for this project. Investigators still need to relate events across entities and time, distinguish plausible links from unsupported assumptions, and identify the assets involved.

The project therefore proposes an AI-assisted platform combining heterogeneous log processing, anomaly detection, entity correlation, and temporal analysis. The system should present a probable incident sequence with traceable evidence and make uncertainty visible so that an investigator can validate or reject the interpretation.

2.2 Objectives

1. To collect authentication, system, network, file, and process logs for a unified investigation workflow.
2. To preprocess and normalize heterogeneous records while retaining links to their original evidence.
3. To extract behavioural and temporal features and identify anomalous activity using AI-assisted analysis.
4. To support the categorization of events as normal, suspicious, or potentially malicious without equating an anomaly with a confirmed attack.
5. To correlate events across users, devices, IP addresses, resources, and timestamps.
6. To reconstruct a probable incident sequence and identify potentially affected assets.
7. To present an evidence-linked timeline, an attack-path graph, and risk and confidence indicators through an investigation dashboard.
8. To keep final incident conclusions subject to human validation.

Chapter 3

LITERATURE SURVEY

The literature survey follows the ten papers selected in the project presentation, focusing on log anomaly detection, semantic representation, event relationships, and cross-system learning. LogDLR is the selected base paper [1]. The descriptions of the other studies follow the supplied presentation, with publication details supplemented from the reference list of the supplied base paper. The limitations distinguish the scope of these detection methods from the broader reconstruction workflow proposed for this project.

3.1 [1] LogDLR: Selected Base Paper

Authors: Junwei Zhou, Shaowen Ying, Shulan Wang, Dongdong Zhao, Jianwen Xiang, Kaitai Liang, and Peng Liu

Published: 2025

About: LogDLR proposes unsupervised cross-system log anomaly detection through domain-invariant latent representations. It addresses differences in log formats and syntax that make system-specific detection models difficult to transfer [1].

Methodology: Log entries are preprocessed without extracting log templates and encoded using SBERT. A Transformer autoencoder learns representations of normal sequences from source and target systems. A domain discriminator and gradient reversal layer support domain-adversarial training so that the encoder learns shared representations. Sequences are flagged when their reconstruction errors exceed a validation-selected threshold.

Advantages: The method avoids dependence on a log-template parser and supports transfer across heterogeneous systems. Model training does not require labelled anomalies. In the paper's cross-system experiments on Thunderbird, Spirit, and BGL, the mean target-system F1-score is 0.930. This is a published result of LogDLR, not a measured result of the proposed project.

Limitations: Evaluation is limited to the selected benchmark systems, and anomalies contaminating the training data can reduce detection quality. Although model training is unsupervised, the reported threshold-selection procedure uses labelled validation data. The paper addresses anomaly detection rather than evidence-linked forensic timelines or complete incident reconstruction.

3.2 [2] LogGraph: Fine-Grained Anomaly Diagnosis

Authors: Jiangming Li, Huasen He, Shuangwu Chen, Dong Jin, and Jian Yang

Published: 2024

About: LogGraph studies robust fine-grained anomaly diagnosis through log event graph learn-

ing and examines relationships among events [2].

Methodology: Association rules separate concurrent log sequences. Events become graph nodes with semantic embeddings, and graph connections represent their relationships. An attention-based gated graph neural network learns the graph context for anomaly diagnosis. The presentation lists HDFS and Network Switch as evaluation datasets.

Advantages: The graph representation supports analysis of noisy and dynamic logs and concurrent anomalies. It provides finer-grained diagnostic information than a single sequence-level anomaly label.

Limitations: Association-rule processing and graph construction introduce complexity. The method focuses on log anomaly diagnosis rather than reconstructing a complete cyber incident across multiple forensic evidence sources.

3.3 [3] SwissLog: Detection and Localization

Authors: Xiaoyun Li, Pengfei Chen, Linxiao Jing, Zilong He, and Guangba Yu

Published: 2023

About: SwissLog addresses anomaly detection and localization in interleaved, unstructured logs from distributed components [3].

Methodology: ID relation graphs group interleaved logs across components. An online parameter-free parser is followed by semantic and temporal embeddings. An attention-based bidirectional long short-term memory network performs detection, and heuristic methods support localization. The presentation lists HDFS, BGL, Android, and Hadoop-blk datasets.

Advantages: The method handles interleaved logs, avoids manual parser-parameter tuning, and supports instance-level anomaly localization. Its use of identifiers and temporal information is relevant to the project's interest in related events.

Limitations: Localization depends on heuristics, and the focus remains log anomaly analysis. It does not provide the proposed full incident reconstruction workflow across heterogeneous forensic sources.

3.4 [4] LogRep: Semantic and Numeric Information

Authors: Xiaoda Xie, Songlei Jian, Chenlin Huang, Fengyuan Yu, and Yunjia Deng

Published: 2023

About: LogRep studies anomaly detection by retaining semantic and numeric information in raw log messages [4].

Methodology: Semantic representations are combined with position-aware numeric representation learning. Attention-based fusion brings the information together for detection. The presentation identifies HDFS and BGL datasets and highlights learning with limited labelled data.

Advantages: Retaining numeric variables and information such as IP-related values can preserve details that template-only representations may omit. The method emphasizes raw-message information rather than only event templates.

Limitations: Performance depends on representation quality and the available training data. Evaluation focuses on benchmark log anomaly detection rather than evidence-linked cyber incident reconstruction.

3.5 [5] DeepUserLog: User-Log Anomaly Detection

Authors: Wenjing Wang, Shida Lu, Jianhui Luo, and Chengrong Wu

Published: 2023

About: DeepUserLog examines anomalous user-log behaviour using semantic analysis and key-value data [5].

Methodology: Key-value pairs are extracted without heavy parsing. Semantic vectors are produced after removing values, while numeric and textual values are analysed separately. The presentation describes SimCSE/Transformer-based representations and temporal information, with evaluation on HDFS, BGL, and LifeUserLog.

Advantages: The method supports complex user logs and incorporates user behaviour and time. Separating message semantics from associated values helps retain contextual information.

Limitations: The approach is oriented toward user-log analysis and depends on suitable key-value information. It does not itself provide the full multi-source correlation and reconstruction workflow proposed here.

3.6 [6] Pre-Training Hierarchical Transformers

Authors: Shaohan Huang, Yi Liu, Carol Fung, He Wang, Hailong Yang, and Zhongzhi Luan

Published: 2023

About: This study improves log-based anomaly detection by pre-training hierarchical Transformers to represent entries and their sequence context [6].

Methodology: Logs are converted into templates. Hierarchical bidirectional Transformers are pre-trained at the log-entry and sequence levels, and the representations are fine-tuned for anomaly detection. The presentation lists HDFS and BGL datasets.

Advantages: The hierarchy captures message-level and sequence-level information. The presentation highlights contextual learning and robustness to unstable logs.

Limitations: The approach depends on log parsing, and pre-training and fine-tuning require computational resources. It addresses anomaly detection rather than forensic reconstruction or an evidence-linked timeline.

3.7 [7] DeepSyslog: Embeddings and Metadata

Authors: Junwei Zhou, Yijia Qian, Qingtian Zou, Peng Liu, and Jianwen Xiang

Published: 2022

About: DeepSyslog studies deep anomaly detection in Syslog using sentence embeddings and event metadata [7].

Methodology: Sentence embeddings capture semantic and contextual information. Metadata

is incorporated to identify anomalies associated with correlated events or exceptional metadata values. The presentation lists HDFS, BGL, and OpenStack datasets.

Advantages: Contextual representations and metadata broaden the information available for analysis. The presentation reports benefits in distinguishing unusual event contexts and reducing false alarms.

Limitations: The method primarily addresses log-stream analysis. It does not generate the proposed evidence-linked forensic timeline or complete multi-source reconstruction of an incident.

3.8 [8] LogBERT: Log Anomaly Detection via BERT

Authors: Haixuan Guo, Shuhan Yuan, and Xintao Wu

Published: 2021

About: LogBERT uses BERT-based self-supervised learning to identify deviations from normal log behaviour [8].

Methodology: The model learns from normal sequences using masked log-message prediction and a hypersphere-based objective. Anomalies are identified through deviations from learned normal patterns. The presentation lists HDFS, BGL, and Thunderbird datasets.

Advantages: The method learns without labelled anomalies and uses bidirectional context to capture dependencies within sequences.

Limitations: BERT-based processing requires computational resources, and detection depends on the learned representation of normal behaviour. The method does not reconstruct attack sequences or establish relationships across all proposed forensic evidence sources.

3.9 [9] NeuralLog: Detection Without Log Parsing

Authors: Van-Hoang Le and Hongyu Zhang

Published: 2021

About: NeuralLog investigates log anomaly detection without a conventional log-template parsing stage [9].

Methodology: Raw messages are converted into semantic vectors and processed by a Transformer with multi-head self-attention to model sequence context. The presentation highlights handling of previously unseen words and lists HDFS, BGL, Thunderbird, and Spirit datasets.

Advantages: Avoiding template parsing reduces dependence on parser quality and preserves raw-message information for semantic analysis.

Limitations: Transformer processing introduces computational requirements. The method focuses on anomaly classification rather than multi-source evidence correlation, forensic explanations, or complete reconstruction.

3.10 [10] LogTAD: Cross-System Domain Adaptation

Authors: Xiao Han and Shuhan Yuan

Published: 2021

About: LogTAD studies unsupervised cross-system log anomaly detection through domain adaptation, particularly when target-domain data are limited [10].

Methodology: Adversarial domain adaptation aligns source- and target-system log-sequence distributions. A shared representation transfers information about normal behaviour to the target system. The presentation lists BGL and Thunderbird datasets.

Advantages: The method supports cross-system transfer without labelled anomalies and reduces dependence on extensive target-system data. It provides an earlier domain-adaptation approach against which LogDLR is compared.

Limitations: Transfer depends on learning useful shared patterns. Large differences between systems can make alignment difficult. The method addresses detection rather than the proposed reconstruction and investigator-review workflow.

3.11 Literature Review Summary

Table 3.1: Literature Review Summary

Ref.	Focus	Key Strength	Gap / Limitation
[1]	LogDLR: cross-system anomaly detection	Parser-free SBERT embeddings and domain-adversarial learning; mean target F1 of 0.930	Labelled validation for threshold selection; training contamination risk; no reconstruction
[2]	LogGraph: fine-grained anomaly diagnosis	Event graphs and attention-based graph learning	Graph and association-rule complexity; log-focused diagnosis
[3]	SwissLog: interleaved log analysis	ID relation graphs and instance-level localization	Heuristic localization; no complete reconstruction
[4]	LogRep: raw-message representation	Retains semantic and numeric information	Representation-dependent; benchmark anomaly detection
[5]	DeepUserLog: user-log analysis	Combines semantics, key-value data, and temporal information	Depends on suitable user-log fields; limited multi-source correlation
[6]	Hierarchical Transformer pre-training	Captures entry-level and sequence-level context	Parsing dependency and training resources
[7]	DeepSyslog: contextual log analysis	Combines sentence embeddings and metadata	Log-stream scope; no evidence-linked incident timeline
[8]	LogBERT: self-supervised detection	Learns normal patterns with bidirectional context	Computational requirements; no attack-sequence reconstruction
[9]	NeuralLog: parser-free detection	Preserves raw-message semantics	Transformer resources; classification-oriented
[10]	LogTAD: cross-system transfer	Domain adaptation with limited target data	Difficult alignment across dissimilar systems; detection-focused

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